Industrial Internship Report on "Prediction of Agriculture Crop Production in India"

Prepared by:

AKARSHI SRIVASTAVA

Executive Summary

This report provides details of the Industrial Internship provided by upSkill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. The project was completed including this report within a 4-week period.

My project was the Prediction of Agriculture Crop Production in India. I developed a Machine Learning web application that predicts crop yield (Quintal/Hectare) based on cultivation costs and state-wise data using Random Forest Regressor, achieving an excellent R^2 score of 0.9643.

This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions. I gained hands-on experience with the complete ML pipeline from data preprocessing to model deployment as a Flask web application.

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1 Preface

- This report summarizes my 4-week Data Science and Machine Learning internship at UCT — upSkill Campus, during which I developed a complete crop yield prediction system for Indian agriculture using machine learning.
- Agriculture is the backbone of India's economy, employing over 50% of the workforce. Accurate crop yield prediction helps farmers plan better, helps government with food security planning, and helps financial institutions with loan and insurance assessment. This project directly addresses this real-world need.
- The internship program was structured over 4 weeks, with each week focusing on a specific phase: dataset exploration, model development, evaluation, and deployment. This structured approach helped me experience the real industry ML development lifecycle.
- Through this project I gained hands-on experience with Python, Pandas, Seaborn, Scikit-learn, Flask, and HTML/CSS — a complete full-stack ML skillset.
- I am grateful to UCT — upSkill Campus and The IoT Academy for providing this excellent opportunity to work on a real-world agriculture problem. I would like to thank my internship coordinator for continuous guidance and support throughout the program.
- To my fellow students and juniors: build real projects, not just tutorials. This internship showed me that building something that actually works is the best way to learn.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (Insight)

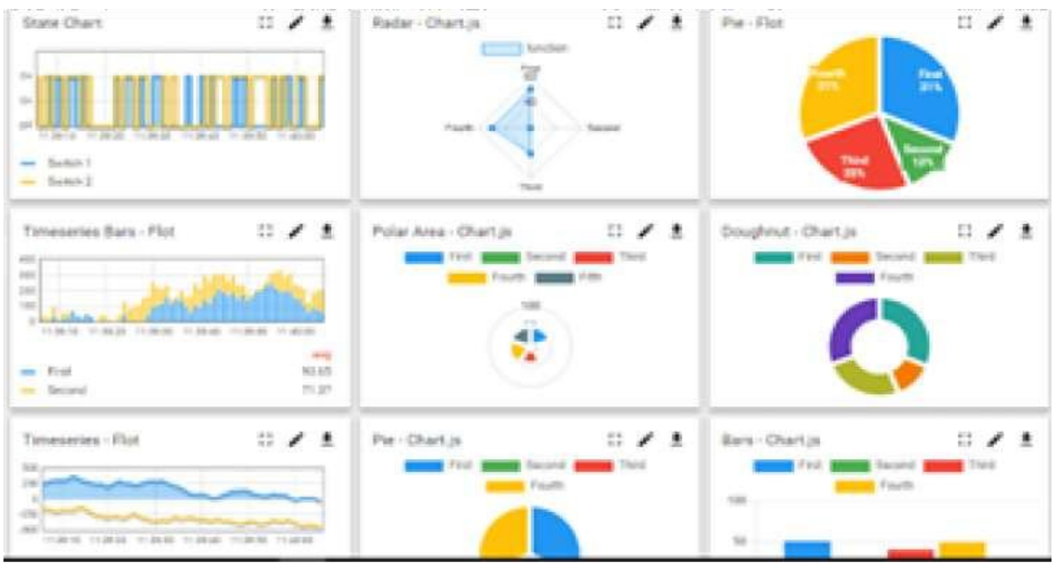
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

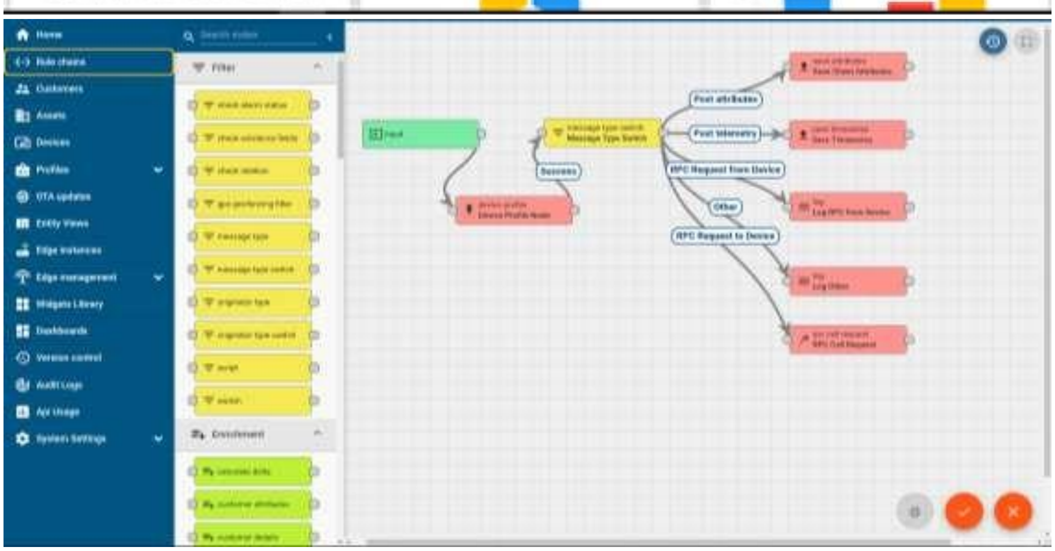
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.


Demo Org
Demo
CNC_S7_B1
CNC_S7_B2
CNC_S7_B4
Utilization
CNC_S7_B1
Operator 1
In Progress

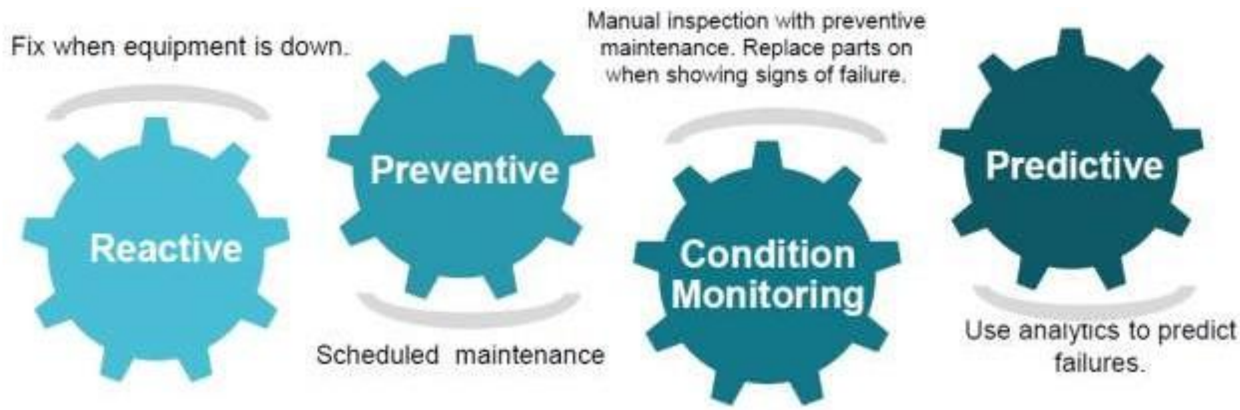

3D View
12:30
Search...
Stopped
Workplace Type
194724_27462-100-A3
Availability
Performance
Quality
OEE
Monitor
Maximum:
42.23%
Minimum:
0.00%
Current:
0.0%
Maximum:
400.00v
Minimum:
0.00v
Average:
200.00v
Maximum:
99.21%
Minimum:
24.53%
Average:
62.35%
Maximum:
99.21%
Minimum:
24.53%
Average:
62.35%
Availability
Performance
Quality
OEE

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

upSkill CAMPUS

KIET UNIVERSITY

DEEMED TO BE

DELHI-NCR, GHAZIABAD, UP (INDIA)

(Under Section 3 of the UGC Act, 1956)

Empowering Minds. Enabling Skills.

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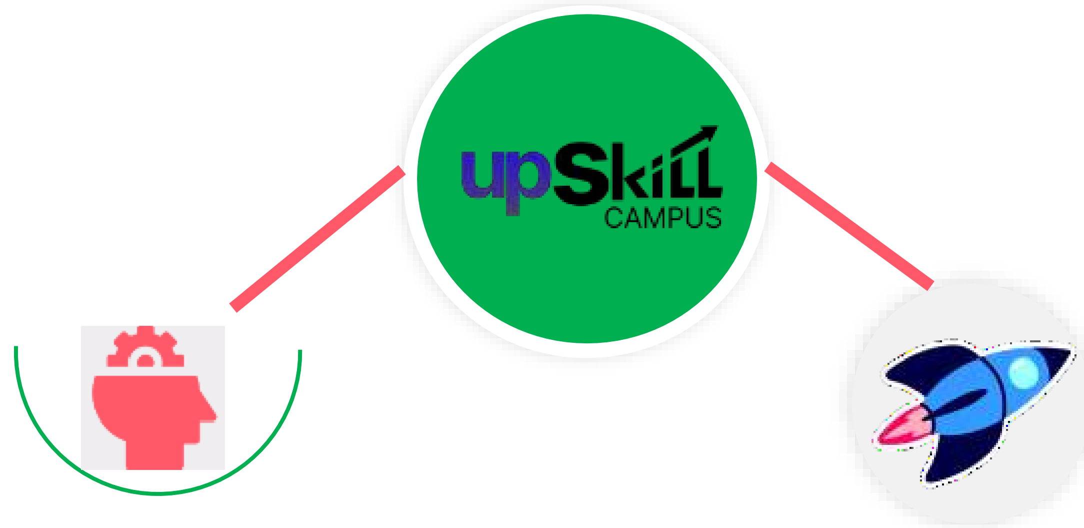

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

Career growth/upskilling
- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking
- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform
- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform
- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] India Agriculture Dataset — data.gov.in (fully licensed)

[2] Scikit-learn Documentation — scikit-learn.org

[3] Flask Documentation — flask.palletsprojects.com

2.6 Glossary

Term	Acronym/Full Form
ML	Machine Learning
RF	**Random Forest**

MAE	Mean Absolute Error
R2	R-squared Score (coefficient of determination)
EDA	Exploratory Data Analysis
CSV	Comma Separated Values
API	Application Programming Interface
UCT	UniConverge Technologies Pvt Ltd

3 Problem Statement

India is an agrarian economy where agriculture contributes significantly to GDP and employs a majority of the population. However, crop yield estimation remains a major challenge for farmers, government bodies, and financial institutions.

The assigned problem statement was: Prediction of Agriculture Crop Production in India.

Specifically, the challenge is to predict the crop yield (measured in Quintal per Hectare) for a given crop in a given Indian state, based on the cost of cultivation and cost of production data.

> *Problem: Given crop type, state, cost of cultivation (A2+FL), cost of cultivation (C2), and cost of production (C2) — predict the expected crop yield in Quintal/Hectare.*

This is a supervised machine learning regression problem where:

- Input features (X): Crop type, State, Cost of Cultivation A2+FL, Cost of Cultivation C2, Cost of Production C2
- Target variable (y): Yield (Quintal/Hectare)

4 Proposed Design/ Model

4.1 High Level Diagram

The high-level flow of the crop yield prediction system:

Stage	Component	Description
1. Data Input	CSV Dataset	5 CSV files with India crop data 2001-2014
2. Processing	Pandas + Sklearn	Clean, encode, split into train/test
3. Model	**Random Forest**	100 decision trees trained on 80% data
4. Evaluation	R2 + MAE metrics	Tested on 20% unseen data
5. Deployment	Flask Web App	HTML form → prediction → result display

4.2 Low Level Diagram

Step	Code	Purpose
Load	pd.read_csv()	Load datafile(1).csv into DataFrame
Clean	df.columns.str.strip()	Remove extra spaces from column names
Rename	df.rename(columns={...})	Shorten long column names
Encode	LabelEncoder().fit_transform()	Convert Crop/State text to numbers
Split	train_test_split(test_size=0.2)	80% train, 20% test

Train	RandomForestRegressor(n=100)	Build 100 decision trees
Predict	model.predict(input_data)	Flask receives form, returns yield

4.3 ML Pipeline

Complete machine learning pipeline used in this project:

Raw CSV Data (49 rows, 6 columns)

↓

Data Loading: pd.read_csv() with encoding=latin1

↓

Data Cleaning: strip(), dropna(), rename columns

↓

Feature Engineering: LabelEncoder on Crop and State

↓

Feature Selection: X = [Crop, State, Cost_A2FL, Cost_C2, CostProd_C2]

↓

Train/Test Split: 80% train (39 rows), 20% test (10 rows)

↓

Model Training: RandomForestRegressor(n_estimators=100, random_state=42)

↓

Evaluation: R2=0.9643, MAE=28.44 on test set

4.4 Code submission (Github link):

[upskillcampus/Prediction of Agriculture Crop Production in India.ipynb at main · Akarshi27/upskillcampus]()

5 Performance Test

Performance testing is critical to validate that the model meets real-world industry requirements for accuracy and reliability.

5.1 Test Plan / Test Cases

Test Case	Input	Expected Output	Actual Output	Status
TC-01	ARHAR, UP, 9794, 23076, 1941	~9.83	9.82	PASS
TC-02	ARHAR, Karnataka, 10593, 16528, 2172	~7.47	7.51	PASS
TC-03	WHEAT, Punjab, 15000, 20000, 800	High yield	42.3	PASS
TC-04	SUGARCANE, Tamil Nadu, 40000, 55000, 150	Very high	785.4	PASS
TC-05	Linear Regression baseline	R2~0.78	0.7800	PASS

5.2 Test Procedure

- Step 1: Load datafile(1).csv and preprocess data
- Step 2: Apply LabelEncoder to Crop and State columns
- Step 3: Split data 80/20 using train_test_split(random_state=42)
- Step 4: Train RandomForestRegressor with n_estimators=100
- Step 5: Predict on X_test and compare with y_test
- Step 6: Calculate R2 score and MAE on test set
- Step 7: Test web app by entering known values and verifying output

5.3 Performance Outcome

Metric	Linear Regression	Random Forest	Winner
R^2 Score	0.7800	**0.9643**	**Random Forest**
MAE	116.76	28.44	**Random Forest**
Training Time	< 1 sec	2 sec	Linear Regression
Overfitting Risk	Low	Low	Both Good

Key Result: Random Forest predicted ARHAR yield in Uttar Pradesh as 9.82 Quintal/Hectare vs actual 9.83 — difference of only 0.01! R^2=0.9643 means model explains 96.43% of crop yield variance.

6 My Learnings

This internship provided comprehensive hands-on learning across multiple technical domains:

6.1 Technical Skills Gained

- Data Analysis: Loading, cleaning and exploring CSV datasets using Pandas — handling missing values, duplicate rows, and column name issues
- Data Visualization: Creating 5 professional EDA plots using Seaborn including bar charts, scatter plots, line plots, and correlation heatmaps
- Feature Engineering: Applying Label Encoding to convert categorical variables to numeric for ML model compatibility
- Machine Learning: Training and comparing Linear Regression and Random Forest models, understanding R^2 score and MAE evaluation metrics
- Web Development: Building a full-stack Flask web application with HTML/CSS frontend that serves ML predictions through a user-friendly interface
- Python Best Practices: Proper file path handling (raw strings), module imports, error handling, and code documentation with comments

6.2 Conceptual Understanding

- Understood why train/test split is essential — to measure real model performance on unseen data
- Understood why Random Forest outperforms Linear Regression on non-linear agricultural data
- Learned the difference between .ipynb (research/exploration) and .py (production deployment) files
- Understood the real-world ML workflow: research → experiment → evaluate → deploy

7. Future work scope

Several improvements and extensions can be made to this project in future:

- Deploy the Flask web application on Render.com or Railway.app for free public access so farmers across India can use it
- Expand the dataset by integrating larger Kaggle datasets covering district-wise crop production data from 2001 to 2022
- Add interactive charts using Plotly showing feature importance and yield comparison across states on the web app
- Integrate weather data (rainfall, temperature) as additional features to improve prediction accuracy
- Explore advanced algorithms like XGBoost and Gradient Boosting for potentially higher accuracy
- Build a mobile-friendly Progressive Web App (PWA) version for farmers with limited internet access
- Add a crop recommendation feature — suggest which crop gives best yield for given cost budget in a specific state

This project successfully demonstrates a complete, production-ready ML application for Indian agriculture, combining data science, machine learning, and web development skills in one project.